

VertexERP AI — Database Schema Specification

Data Model Dictionary, ERD References & Indexing Guide

1. Relational Schema Architecture

VertexERP AI operates over 90+ normalized relational entities in PostgreSQL 17. Every table enforces strict tenant isolation using an indexed organization_id foreign key, standard UUID primary keys, UTC timestamps (created_at, updated_at), and soft delete flag (is_deleted).

2. Core Entity Relationship Specifications

Table Name	Primary Key	Foreign Keys / Indexes	Key Attributes
organizations	id (UUID)	slug (UNIQUE INDEX)	name, slug, currency, status
users	id (UUID)	organization_id, role_id	email (UNIQUE), password_hash, is_active
employees	id (UUID)	organization_id, department_id	employee_code, first_name, designation, salary
customers	id (UUID)	organization_id	customer_code, company_name, email, phone
products	id (UUID)	organization_id, category_id	sku (UNIQUE), name, unit_price, cost_price
customer_invoices	id (UUID)	organization_id, customer_id	invoice_number, total_amount, status, due_date
bill_of_materials	id (UUID)	organization_id, product_id	bom_code, revision, is_active, is_approved
production_orders	id (UUID)	organization_id, bom_id	order_number, quantity, status, start_date
rag_documents	id (UUID)	organization_id, collection_id	title, document_type, storage_path, vector_status
ml_models	id (UUID)	organization_id	model_code, algorithm, accuracy_score, stage

3. Indexing & Optimization Strategy

Composite B-Tree indexes are configured on (organization_id, is_deleted, status) across all transaction tables. Vector embedding tables leverage PgVector HNSW (Hierarchical Navigable Small World) indexes with cosine similarity calculation.